

TBILISI INTL, Georgia

WMO: 375450

Lat: 41.667N

Lon: 44.950E

Elev: 1624

StdP: 13.85

Time Zone: 4.00 (E04)

Period: 05-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.7	23.7	10.2	9.8	23.8	14.3	12.1	28.3	42.1	35.2	38.5	36.7	4.1	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.6	95.2	71.4	93.0	70.8	90.1	70.7	76.7	89.4	74.6	87.0	72.7	84.8	11.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	129.6	85.3	70.7	120.1	82.8	68.7	112.0	80.5	41.5	89.1	39.4	87.3	37.6	85.0	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
37.2	31.9	28.0	16.9	98.5	5.3	3.1	13.1	100.7	10.0	102.5	7.0	104.2	3.1	106.4
			14.8	79.5	5.2	1.7	11.1	80.8	8.0	81.8	5.1	82.7	1.3	84.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.7	36.4	38.5	46.9	55.4	65.2	73.6	78.3	78.7	70.1	60.0	48.1	39.6
	DBStd	16.09	6.13	7.09	6.33	7.11	5.65	4.36	4.61	5.26	5.39	5.10	6.29	5.80
	HDD50	1343	422	323	137	26	0	0	0	0	0	2	106	327
	HDD65	4032	887	742	562	297	65	1	1	0	15	166	508	788
	CDD50	4143	0	1	40	187	472	709	877	890	602	313	48	4
	CDD65	1355	0	0	0	7	72	260	412	425	167	12	0	0
	CDH74	12252	0	0	4	82	496	2073	4017	4425	1098	57	0	0
	CDH80	5200	0	0	0	12	95	770	1839	2157	324	3	0	0
Wind	WSAvg	10.2	9.2	10.2	13.1	11.8	10.7	11.2	11.3	10.6	9.6	8.2	8.3	8.1
Precipitation	PrecAvg	18.9	0.7	0.9	1.2	2.0	3.0	2.8	1.7	1.8	1.3	1.5	1.2	0.8
	PrecMax	32.0	2.0	2.6	3.1	5.3	7.1	10.8	6.1	5.1	5.9	3.9	4.1	2.9
	PrecMin	10.0	0.0	0.0	0.0	0.5	0.2	0.8	0.0	0.1	0.0	0.0	0.1	0.0
	PrecStd	4.4	0.6	0.7	0.7	1.3	1.6	1.9	1.5	1.3	1.2	1.0	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	61.2	71.2	80.7	87.4	93.0	98.3	98.9	92.9	79.0	68.3	60.2
		MCWB	44.9	48.8	53.3	62.4	68.7	71.0	71.8	72.8	69.2	63.3	54.9	49.3
	2%	DB	51.5	56.5	66.1	76.0	82.3	89.9	94.9	96.4	87.5	75.2	64.6	55.1
		MCWB	43.6	46.0	52.2	60.3	66.2	70.2	71.2	71.5	68.5	62.5	54.1	46.8
	5%	DB	48.0	52.4	62.4	72.1	79.2	87.7	92.1	93.7	83.3	72.0	60.4	51.5
		MCWB	41.3	44.1	50.5	58.7	64.4	69.5	71.0	70.9	66.7	60.7	51.9	44.8
	10%	DB	44.8	48.6	57.8	67.9	76.3	84.7	89.6	91.3	80.3	69.5	57.1	48.6
		MCWB	39.7	42.5	48.2	56.6	63.5	69.1	71.4	70.8	66.1	59.2	50.6	43.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	50.1	56.6	64.8	70.5	74.5	78.3	80.4	74.8	66.3	57.6	50.7
		MCDB	52.7	59.2	68.7	76.9	83.6	88.3	92.1	92.8	86.9	74.8	65.6	57.4
	2%	WB	44.4	47.0	53.3	62.1	67.7	72.6	75.8	77.9	71.1	63.7	55.4	48.2
		MCDB	49.1	54.8	63.1	73.2	79.8	86.3	89.7	91.0	82.1	71.7	60.8	53.6
	5%	WB	42.1	45.0	51.0	59.8	66.1	71.3	74.4	75.6	69.0	62.0	53.4	46.0
		MCDB	46.7	51.4	60.1	69.3	77.5	84.6	87.4	87.0	79.4	69.4	57.8	50.6
	10%	WB	40.1	42.9	49.0	57.6	64.4	69.8	72.6	73.2	67.2	60.5	51.6	43.5
		MCDB	44.1	47.8	56.6	66.0	74.2	82.1	84.7	84.3	76.9	66.9	56.3	48.1
Mean Daily Temperature Range	5% DB	MDBR	13.7	14.3	16.9	17.3	18.0	19.4	19.6	20.6	17.8	15.9	14.7	14.3
		MCDBR	18.4	20.8	24.9	24.2	23.4	23.8	23.9	25.0	22.2	20.4	20.8	18.7
		MCWBR	13.1	13.5	14.7	13.5	12.4	11.2	11.9	11.7	11.8	11.1	11.9	12.2
	5% WB	MCDBR	16.4	19.4	21.4	22.3	21.1	22.0	21.6	22.5	19.4	18.4	17.2	16.5
		MCWBR	12.2	12.7	13.4	12.9	11.5	11.2	11.4	12.2	11.6	10.8	10.8	11.1
Clear-Sky Solar Irradiance	taub	0.332	0.369	0.415	0.480	0.449	0.439	0.481	0.475	0.424	0.401	0.353	0.330	
	taud	2.327	2.211	2.118	1.977	2.139	2.226	2.119	2.129	2.254	2.254	2.348	2.347	
	Ebn at Noon	254	260	261	252	264	266	253	251	256	246	244	244	
	Edh at Noon	28	36	44	55	48	44	49	47	39	34	27	25	
All-Sky Solar Radiation	RadAvg	589	864	1105	1385	1685	1970	1923	1761	1371	910	642	512	
	RadStd	69	120	119	188	119	138	157	182	116	101	74	66	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page